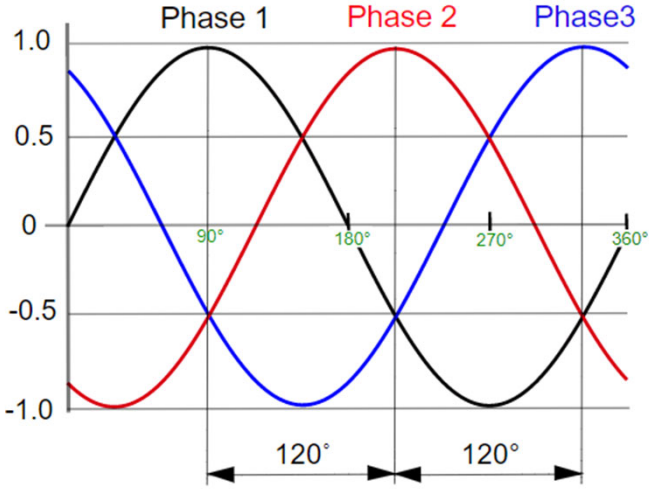
ECE 330

Chapter 2

Phasors and three-phase circuits



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Learning Objectives: you will be able to

- Convert time-domain sinusoidal sources to phasor form using **RMS values**
- Use **RMS phasors** to analyze steady-state AC circuits
- Define and calculate **real (P), reactive (Q)**, **apparent (S)** and **complex power ($\bar{S}$)**
- Define **power factor (PF),** and classify loads as having **leading, lagging, or unity** power factor.
- Use the **power triangle** to compute power and power factor.
- Explain the motivation for **power factor correction** in AC power systems, and calculate the required reactive compensation to achieve a desired PF.
- Explain the motivation for using three-phase power systems.
- Identify **Wye (Y)** and **Delta (Δ)** three-phase connections and analyze balanced three-phase loads connected in Y or Δ configurations.
- Develop **per-phase equivalent circuits** for balanced Y- and Δ-connected loads and use single-phase circuit analysis techniques to solve three-phase power problems

Review of Phasors

- ECE 210

$$A(t) = A_m \cos(\omega t + \theta)$$

magnitude; phase shift

⇓ phasors

$$A = A_m \angle \theta$$

- ECE 330

$$A(t) = A_m \cos(\omega t + \theta)$$

⇓ phasors

$$\bar{A} = \frac{A_m}{\sqrt{2}} \angle \theta$$

A_{rms}

Converting from sine to cosine

$\sin x = \cos(x - 90^\circ)$

$$v(t) = \sqrt{2}(10)\sin\left(wt + 30^\circ\right) =$$

$= \sqrt{2}(10)\cos(wt + 30^\circ - 90^\circ)$

-60°

⇓ phasors

$\bar{V} = \frac{\sqrt{2}(10)}{\sqrt{2}} \angle -60^\circ$ Volts

fix it by $\pm 180^\circ$ shift

$$i(t) = -\sqrt{2}(5)\sin\left(wt - 20^\circ\right) =$$

$= \sqrt{2}(5)\cos(wt - 20^\circ - 90^\circ + 180^\circ)$

70°

⇓ phasors

$\bar{I} = 5 \angle 70^\circ$ Amps

Example 1: write as phasors

$$i(t) = \sqrt{2}(5)\cos(wt - 20^\circ)$$

A) $\bar{I} = \sqrt{2}(5)\angle -20^\circ$

B) $\bar{I} = (5)\angle -110^\circ$

C) $\bar{I} = \sqrt{2}(5)\angle -110^\circ$

D) $\bar{I} = (5)\angle -20^\circ$

Example 2: write as phasors

$$i(t) = -\sqrt{2}(5)\sin\left(wt + 30^\circ\right) = 5\cos\left(wt + 30^\circ - 90^\circ + 180^\circ\right)$$

$\sqrt{2}$ — 120°

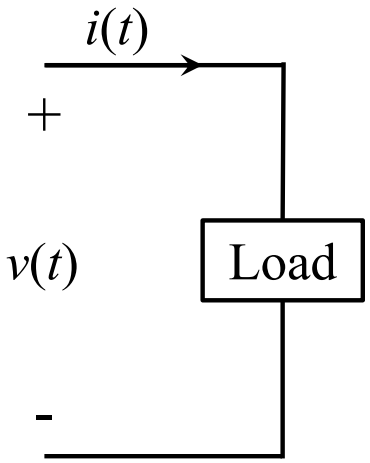
A) $\bar{I} = (5)\angle 120^\circ$

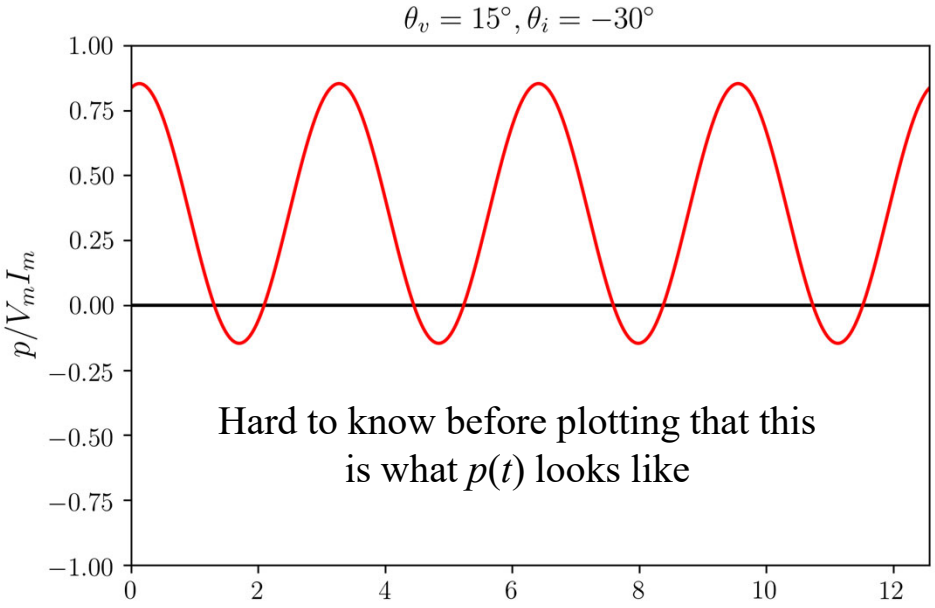
B) $\bar{I} = \sqrt{2}(5)\angle 30^\circ$

C) $\bar{I} = \sqrt{2}(5)\angle 120^\circ$

D) $\bar{I} = (5)\angle 30^\circ$

Single Phase: 1 voltage and 1 current





- $p(t) = v(t)\,i(t)$
- $v(t) = V_m \cos(\omega t + \theta_v)$

  $i(t) = I_m \cos(\omega t + \theta_i)$
- $p(t) = V_m I_m \cos(\omega t + \theta_v)\cos(\omega t + \theta_i)$

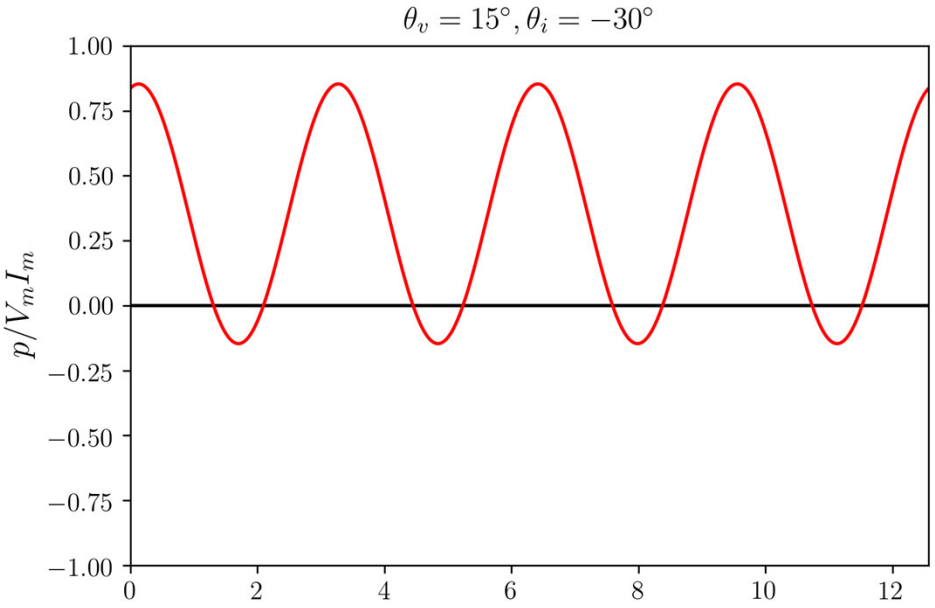
Simplifying p(t)

$$\cos(A)\cos(B) = \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right]$$

$$p(t) = \frac{V_m I_m}{2}\left(\cos(2\omega t + \theta_v + \theta_i) + \cos(\theta_v - \theta_i)\right)$$

$\frac{V_m I_m}{2} = V_{rms} I_{rms}$

- $V_{rms} = \frac{V_m}{\sqrt{2}}$
- $I_{rms} = \frac{I_m}{\sqrt{2}}$



Average Power: P

$$P = \frac{1}{T}\int_0^T p(t)\,dt = \frac{1}{T}\int_0^T V_{rms} I_{rms}\left(\cos(2\omega t + \theta_v + \theta_i) + \cos(\theta_v - \theta_i)\right)dt =$$

$$= \frac{1}{T}\cdot T\cos(\theta_v - \theta_i)\, V_{rms} I_{rms}$$

$$\boxed{P = V_{rms} I_{rms}\cos(\theta_v - \theta_i)}$$

$[W]$

- average (real, active) power "doing the work"

Reactive (Q) and Complex ($\bar{S}$) power

- $P = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$ – real power [W]
- $Q = V_{rms} I_{rms} \sin(\theta_v - \theta_i)$ – reactive power [VAr]

voltamperes reactive

$\Downarrow$

- $P = \mathrm{Re}\{V_{rms} I_{rms} e^{j(\theta_v - \theta_i)}\}$
- $Q = \mathrm{Im}\{V_{rms} I_{rms} e^{j(\theta_v - \theta_i)}\}$

$\Downarrow$

- $P = \mathrm{Re}\{\underbrace{V_{rms} e^{j\theta_v}}_{\bar{V}} \cdot \underbrace{I_{rms} e^{-j\theta_i}}_{\bar{I}^*}\} = \boxed{P = \mathrm{Re}\{\bar{V} \cdot \bar{I}^*\}}$
- $Q = \mathrm{Im}\{V_{rms} e^{j\theta_v} \cdot I_{rms} e^{-j\theta_i}\} = \boxed{Q = \mathrm{Im}\{\bar{V} \cdot \bar{I}^*\}}$

apparent power: $S = V \cdot I$

$S \angle \theta$

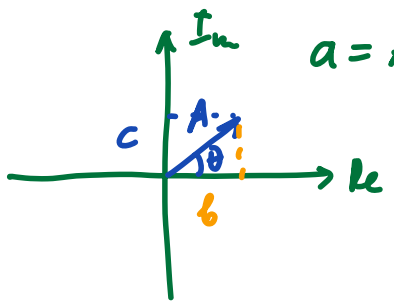
$\boxed{\bar{S} = \bar{V} \cdot \bar{I}^* = P + jQ}$ complex power [VA]

voltamperes

$(\theta_v - \theta_i)$: power factor angle

$\cos(\theta)$: power factor (PF) 0 – 1

Detour



$a = Ae^{j\theta}$

$b = \mathrm{Re}\{Ae^{j\theta}\} = A\cos\theta$

$c = \mathrm{Im}\{Ae^{j\theta}\} = A\sin\theta$

Example 3:

$\bar{V} = 10\angle 15^\circ$ V, $\bar{I} = 5\angle -30^\circ$ A

Find P:

$P = \mathrm{Re}\{\bar{V}\cdot\bar{I}^*\} = \mathrm{Re}\{10\angle 15^\circ \cdot 5\angle 30^\circ\} = \mathrm{Re}\{50\angle 45^\circ\} = 35.35$ W

units, please ☺

Example 4:

$\bar{V} = 10\angle 15^\circ$ V, $\bar{I} = 5\angle -30^\circ$ A

Find Q:

A) 50 W

B) 35.355 W

C) -12.94 VAR

D) 35.355 VAR

Example 5:

$v(t) = \sqrt{2}(10)\cos(\omega t + 15^\circ)$ ⇒ $\bar{V} = 10\angle 15^\circ$

$i(t) = \sqrt{2}(7)\sin(\omega t + 75^\circ)$ ⇒ $\bar{I} = 7\angle -15^\circ$

$= \sqrt{2}(7)\cos(\omega t + 75^\circ - 90^\circ)$

Find $\bar{S}$:

A) $70 + j0$ VA

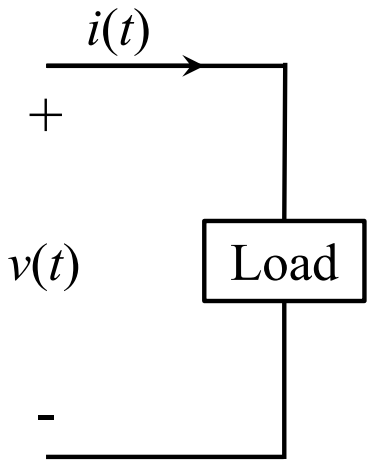
B) $35 - j60.62$ VA

C) $60.62 + j35$ VA

D) $121.24 + j70$ VA

① $\bar{S} = \bar{V} \cdot \bar{I}^* = 10\angle 15^\circ \cdot 7\angle 15^\circ = 70\angle 30^\circ$ VA

Another way to view complex power



$\bar{Z} = R + jX$ ← reactance

↑ resistance

① $\bar{S} = \bar{V} \cdot \bar{I}^* = \bar{Z}(\bar{I} \cdot \bar{I}^*) = \bar{Z} \cdot I^2$

↑ $\bar{V} = \bar{Z} \cdot \bar{I}$

② $\bar{S} = I^2(R + jX) = \underbrace{I^2 \cdot R}_{P} + j\underbrace{I^2 \cdot X}_{Q} = P + jQ$

P: power dissipated by resistors

Q: power generated or consumed by reactances

Note: we can show that $\theta = \theta_z$

$\bar{S} = \bar{Z} \cdot I^2 = I^2 z \angle\theta_z = S\angle\theta \Rightarrow \theta = \theta_z$

cap: $X_C = -\frac{1}{\omega C}$

$Q < 0$

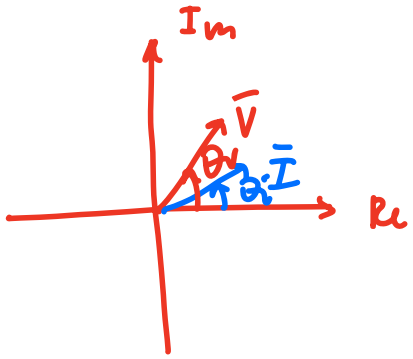
ind: $X_L = \omega L$

$Q > 0$

Recall: cosine is an even function

$$\cos(x)=\cos(-x)$$

PF = cos($\underbrace{\theta_v-\theta_i}_{\theta}$)



Current lags the voltage

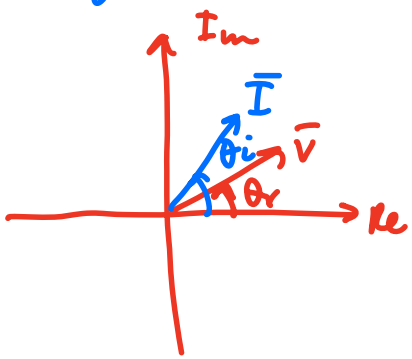
$\theta_v > \theta_i$

$\theta = \theta_v - \theta_i > 0$

Lagging PF

Note: voltage is a reference!

Note: PF is only defined for θ between -90° and 90°



current leads the voltage

$\theta_v < \theta_i$

$\theta < 0$

Leading PF

How do leading and lagging PF correspond to circuit?

Inductor

- $\bar{S}_L$ - ?
- $\bar{S}_L = \bar{V} \cdot \bar{I}^*$

$i(t) = I_m \cos(\omega t + \theta_i) \Rightarrow \bar{I} = I \angle \theta_i$

$v(t) = L \frac{di}{dt} \Rightarrow \bar{V} = L(j\omega \bar{I}) = \omega L I \angle \theta_i + 90^\circ$

$j = 1 \angle 90^\circ$

* $\frac{di}{dt} \rightarrow j\omega \bar{I}$

- $\bar{S}_L = \bar{V} \cdot \bar{I}^* = \omega L I^2 \angle \theta_i + 90^\circ - \theta_i$ (>0)

PF = $\cos(\theta) = 0$ lag

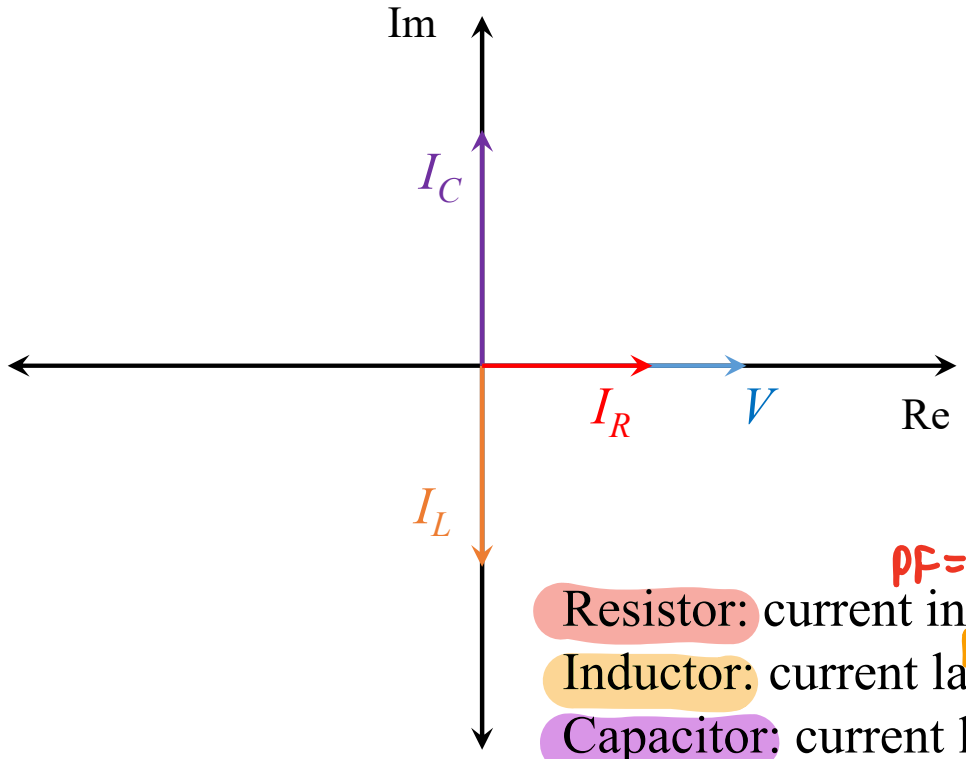
Capacitor

- $\bar{S}_C$ - ?
- $\bar{S}_C = \bar{V} \cdot \bar{I}^*$
- $v(t) = V_m \cos(\omega t + \theta_v) \Rightarrow \bar{V} = V \angle \theta_v$
- $i(t) = C \frac{dv}{dt} \Rightarrow \bar{I} = j\omega \bar{V} C$

- $\bar{S}_C = C\omega V^2 \angle \theta_v - \theta_v - 90^\circ$ (<0)

PF = 0 lead

Plotting in Complex Plane



PF = 1 (unity) Q = 0

Resistor: current in phase with voltage

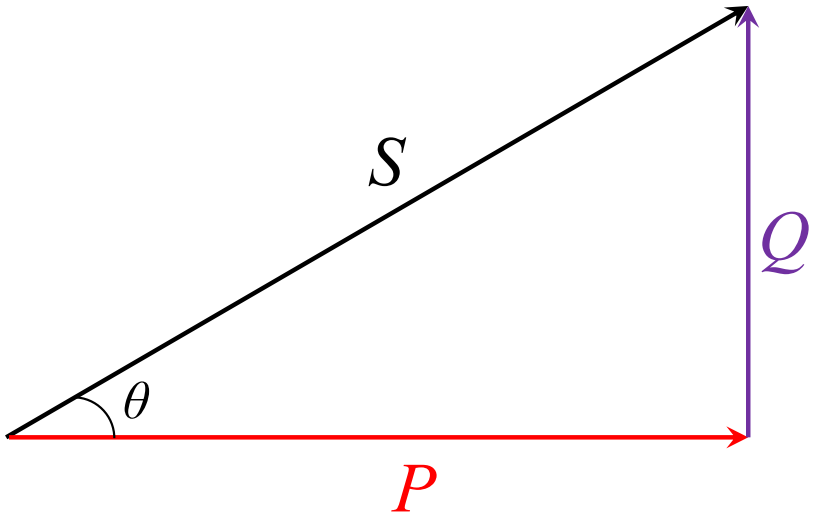
PF = 0 lag P = 0

Inductor: current lags voltage by 90°

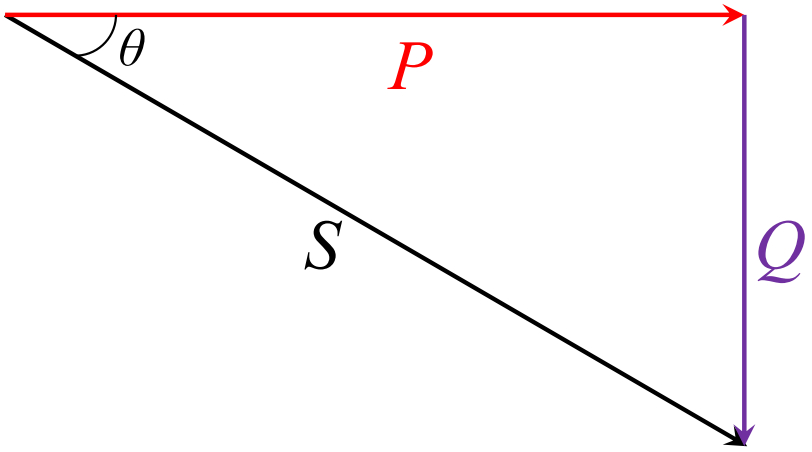
Capacitor: current leads voltage by 90°

PF = 0 lead P −

Power Triangle: Lagging PF



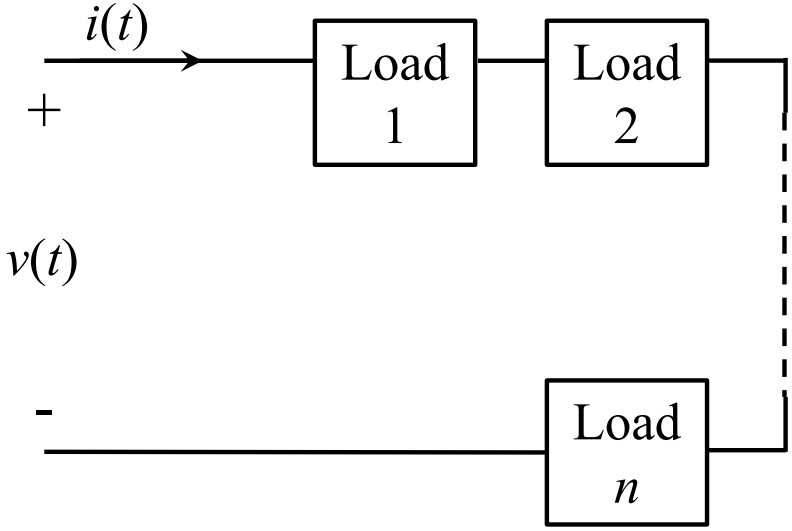
Power Triangle: Leading PF

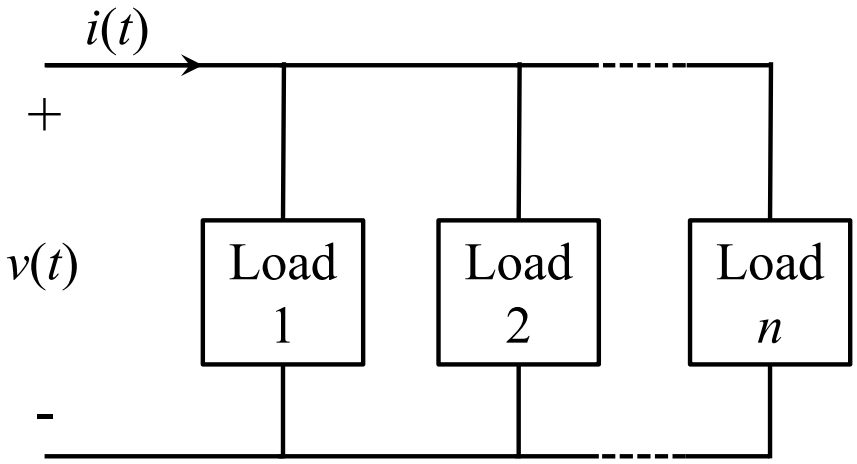


Specifying Power

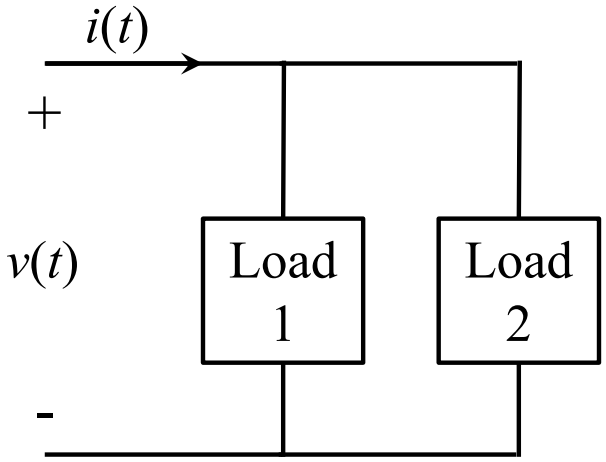
- 6 total quantities:
- Need a reference: assume
- Use whatever is given to calculate unknown quantities

What if there are multiple loads?





Example 6:



$$\bar{V} = 100\angle 10^\circ \text{ V}$$

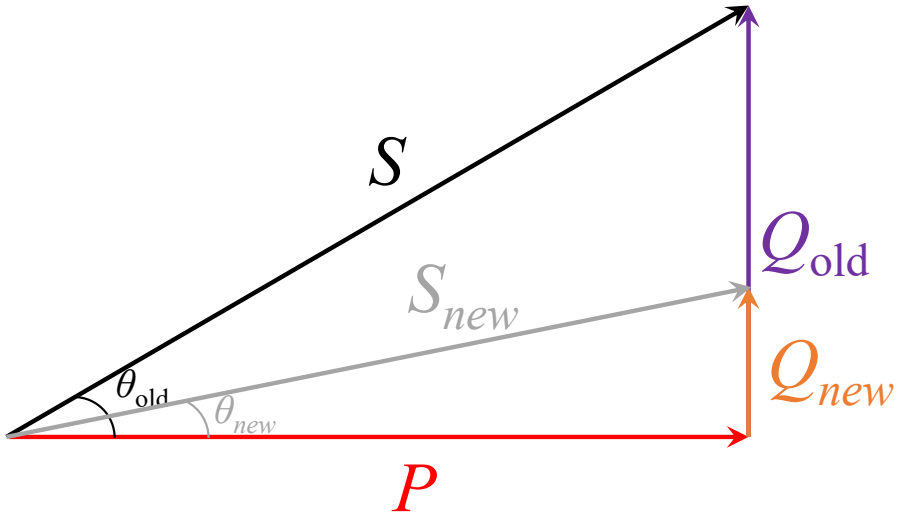
$$\bar{I}_1 = 10\angle -26.8^\circ \text{ A}$$

$$\bar{I}_2 = 5\angle 40^\circ \text{ A}$$

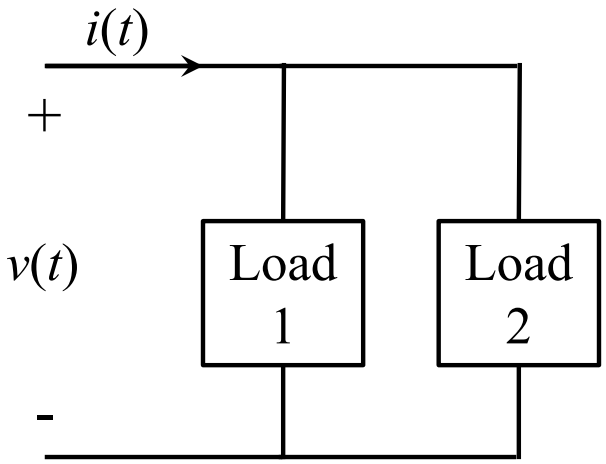
Purely reactive loads:

Why PF correction?

Power Triangle: Power Factor Correction



Example 7: PF correction



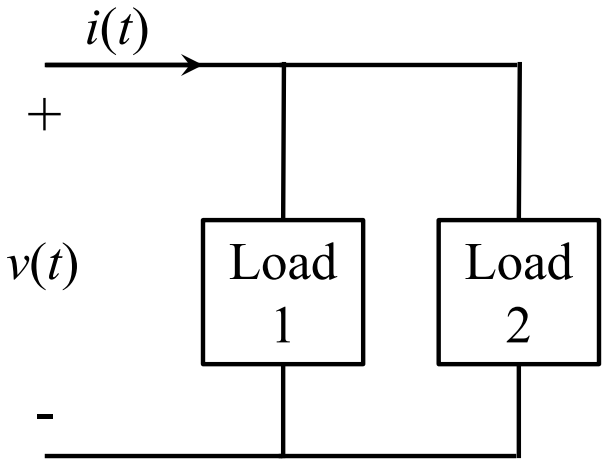
$$\bar{V} = 100\angle 10^\circ \text{ V}$$

$$\bar{I}_1 = 10\angle -26.8^\circ \text{ A}$$

$$\bar{I}_2 = 5\angle 40^\circ \text{ A}$$

What should capacitor in parallel add to get PF = 0.98 lagging?

Example 8: PF correction



$\bar{V} = 100\angle 10^\circ \text{ V}$

$\bar{I}_1 = 10\angle -26.8^\circ \text{ A}$

$\bar{I}_2 = 5\angle 40^\circ \text{ A}$

What should capacitor in parallel add to get PF=1?

A) 0 VAR of capacitance

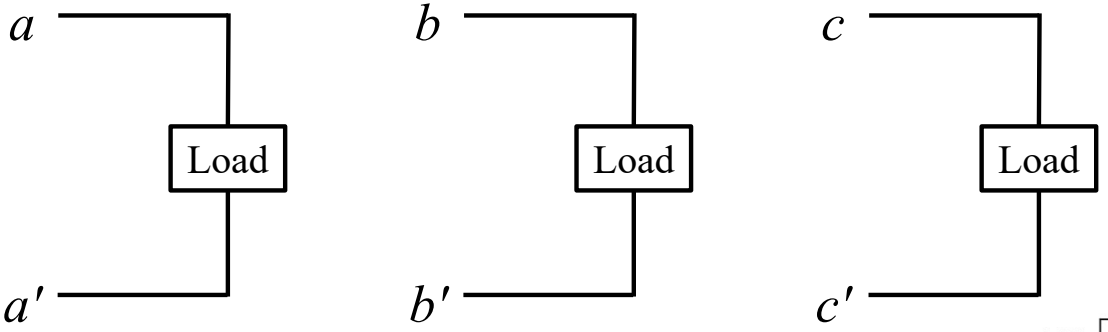
B) 600 VAR of capacitance

C) 100 VAR of capacitance

D) 350 VAR of capacitance

2 Phase system

3 phase power



Positive Sequence:

a - b - c

$$V_{aa'} = V_m \cos(\omega t)$$

$$V_{bb'} = V_m \cos(\omega t - 120^\circ)$$

$$V_{cc'} = V_m \cos(\omega t + 120^\circ)$$

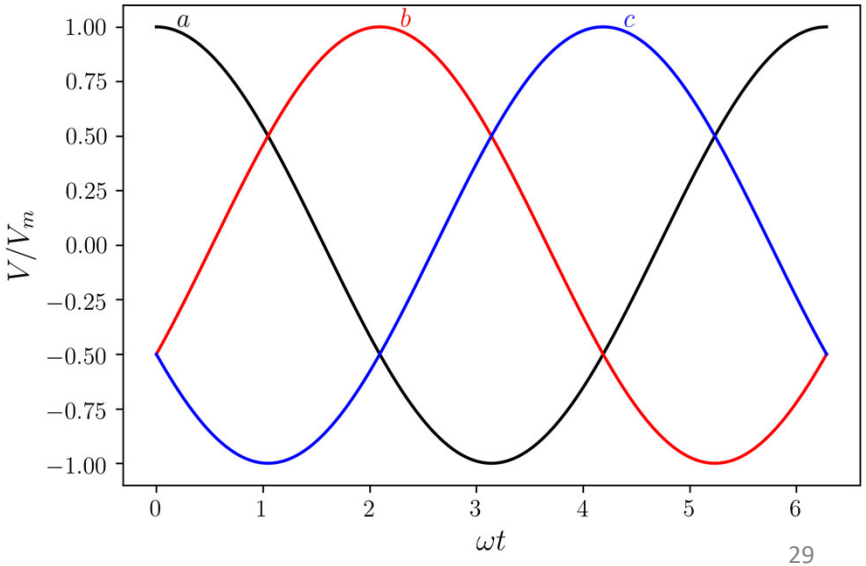
Negative Sequence:

a - c - b

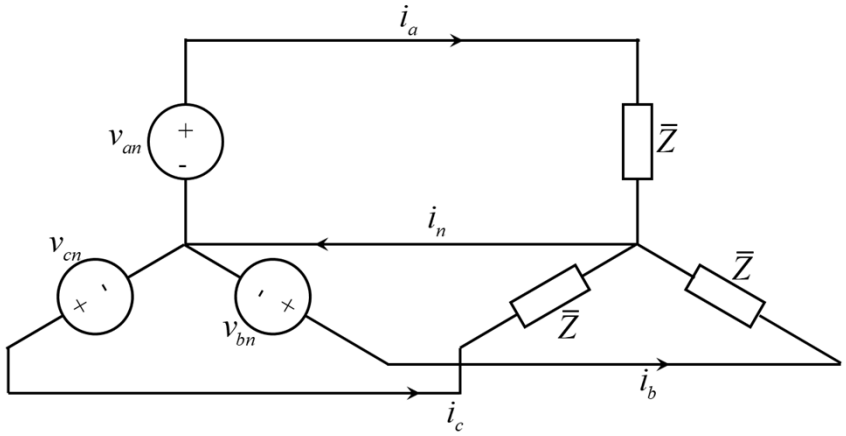
$$V_{aa'} = V_m \cos(\omega t)$$

$$V_{bb'} = V_m \cos(\omega t + 120^\circ)$$

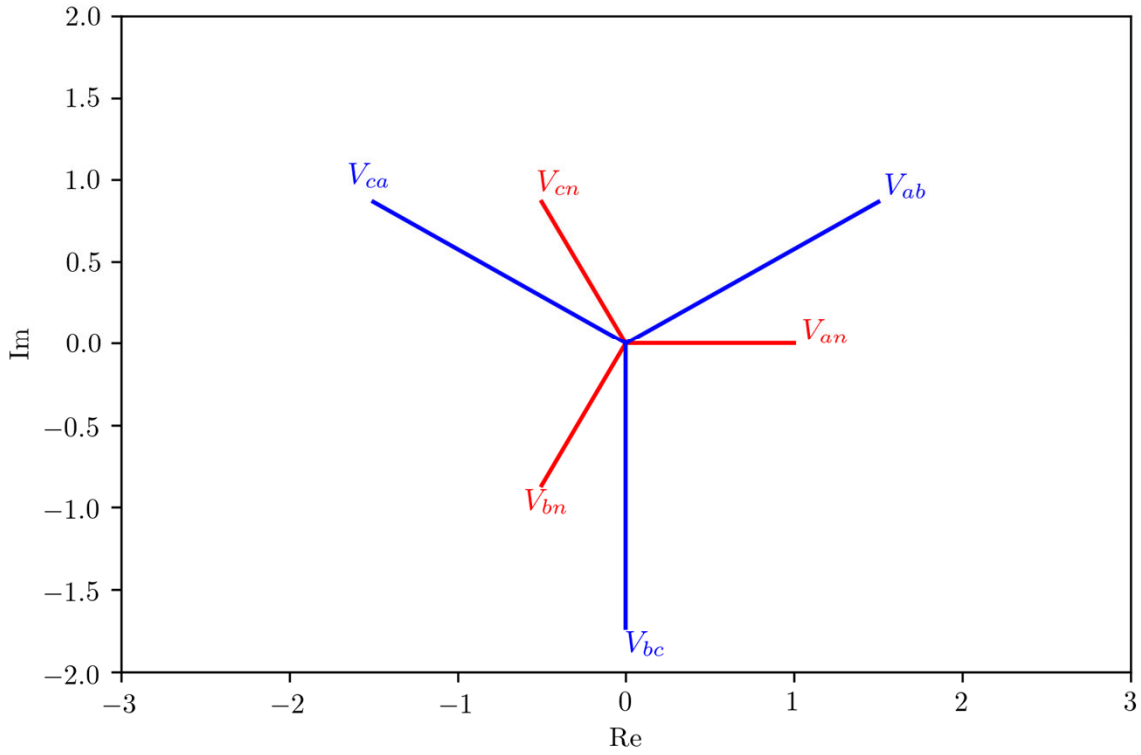
$$V_{cc'} = V_m \cos(\omega t - 120^\circ)$$



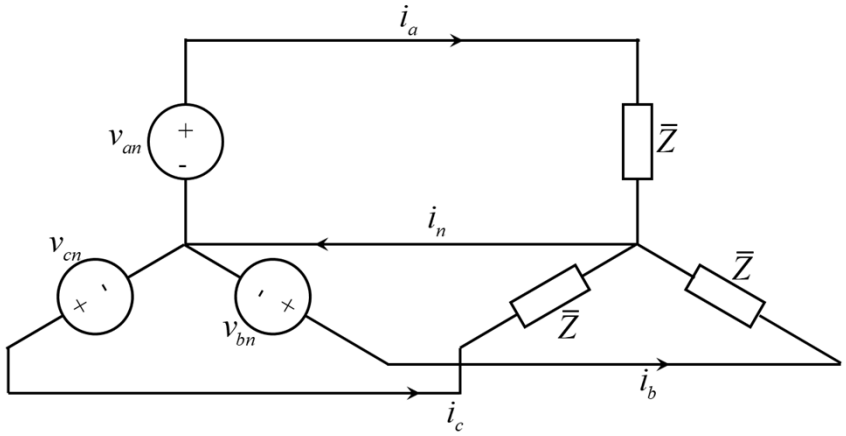
Wye-Wye Connection



Phasor Diagram



Currents in each phase/line



Complex Power

Example 9:

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

Example 9-cont:

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

Wye-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find $\bar{S}$

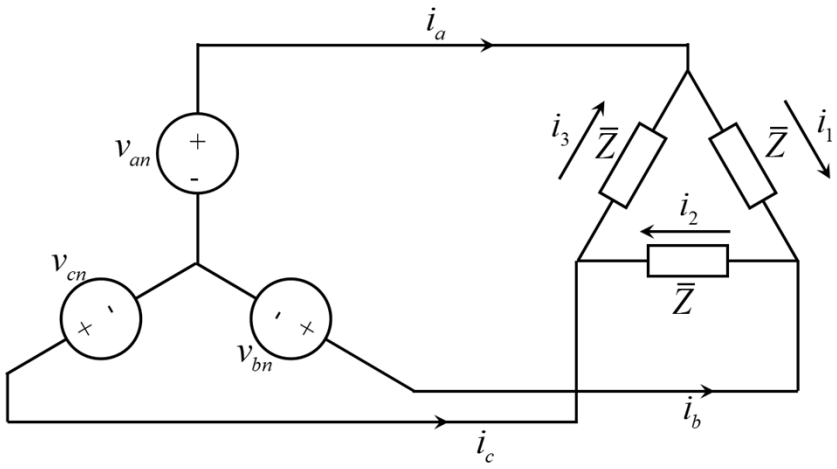
A) $17318.4\angle 36.87^\circ$ VA

B) $30000\angle 36.87^\circ$ VA

C) $30000\angle -36.87^\circ$ VA

D) $17318.4\angle -36.87^\circ$ VA

Wye-Delta Connection



Complex Power

Example 10:

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

A) 50 A

B) 62.5 A

C) 36.08 A

D) 28.87 A

Example 10-cont:

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find I_L

Delta-connected load, $V_L = 480\ V, P = 24\ kW, PF = 0.8\ lag.$
Find $\bar{S}$

A) $17318.4\angle 36.87^\circ$ VA

B) $30000\angle 36.87^\circ$ VA

C) $30000\angle -36.87^\circ$ VA

D) $17318.4\angle -36.87^\circ$ VA

Example 11:

$S_{tot} = 1000k$Va, PF=0.8 lag, $V_L = 4160$ V

Wye load: pure resistors, $P_1 = 200\ kW$

Delta load: impedence $\bar{Z}_\Delta$

Find $\bar{S}_2$:

Example 11-cont

$S_{tot} = 1000k$Va, PF=0.8 lag, $V_L = 4160$ V

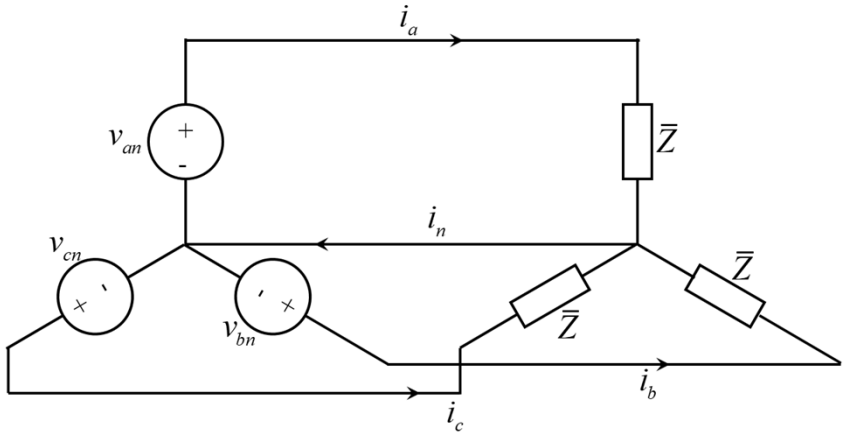
Wye load: pure resistors, $P_1 = 200\ kW$

Delta load: impedence

Find $\bar{S}_2$:

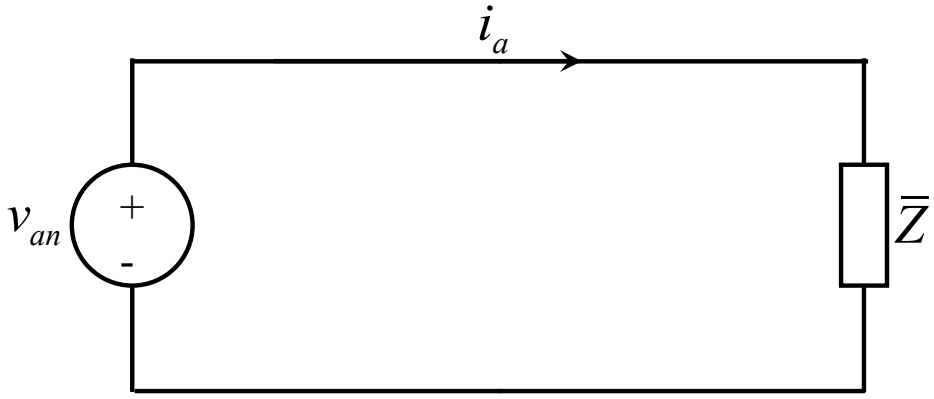
Find $\bar{Z}_\Delta$:

Per phase equivalent circuit: Wye-Wye Connection

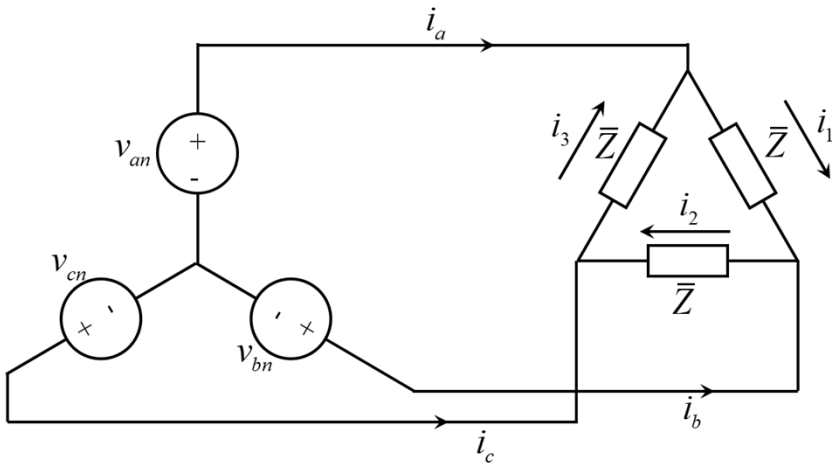


Per phase equivalent circuit

- Look only at a phase



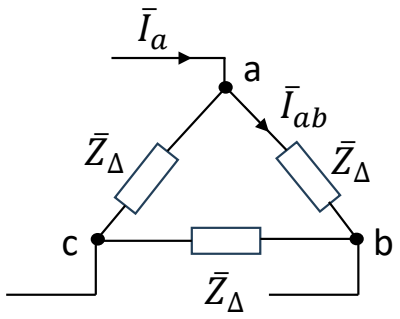
Per phase equivalent circuit: Wye-Delta Connection



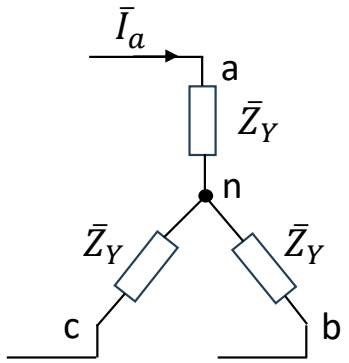
Again, only look at a phase

Delta-Wye Conversion

Delta



Wye



$$\bar{S}_{1\emptyset} = \bar{V}_{ab} \cdot \bar{I}_{ab}^{\,*} = (\bar{Z}_\Delta \cdot \bar{I}_{ab}) \cdot \bar{I}_{ab}^{\,*} = \bar{Z}_\Delta \cdot I_{ab}^{\,2} = \bar{Z}_\Delta \cdot \left(\frac{I_L}{\sqrt{3}}\right)^2$$

$\bar{Z}_\Delta \cdot \bar{I}_{ab}$ ↑

$\frac{I_a}{\sqrt{3}}$ ↑

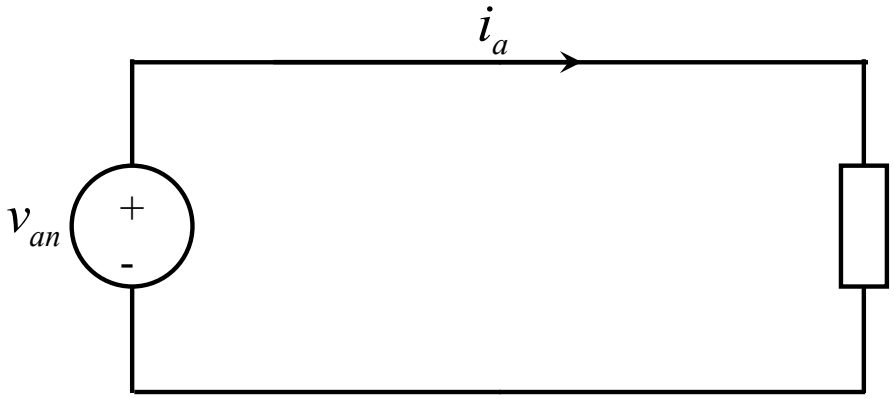
$$\bar{S}_{1\emptyset} = \bar{V}_{an} \cdot \bar{I}_a^{\,*} = (\bar{Z}_Y \cdot \bar{I}_a) \cdot \bar{I}_a^{\,*} = \bar{Z}_Y \cdot I_a^{\,2}$$

$\bar{Z}_Y \cdot \bar{I}_a$ ↑

$$\bar{S}_{1\emptyset,\Delta} = \frac{\bar{Z}_\Delta}{3} \cdot I_L^{\,2} \quad \Rightarrow \quad \bar{Z}_Y = \frac{\bar{Z}_\Delta}{3} \quad \Leftarrow \quad \bar{S}_{1\emptyset,Y} = \bar{Z}_Y \cdot I_L^{\,2}$$

To keep the same line current and complex power

Per phase equivalent circuit



Example 12:

$V_L = 208$ V, 2 Loads in Parallel

Wye load: $P_1 = 300\ kW$, PF=0.95 lag

Delta load: $\bar{Z}_\Delta = 40 + j30\ \Omega$

Find I_L, S_{tot}:

Bonus: PF correction

Given: PF_old = 0.95 lag
S_tot = 318 kVA
We need to get PF_new = 0.98 lag.
Find Q_c.